

CyberQuant: Full Project Blueprint, Status & Execution Plan

AI-Powered Continuous Cyber Risk Quantification & Investment Optimization Platform

Overall Status: ~25% Completed | Foundation Layer 100% Done & Synchronized on GitHub | Milestones 2–6 Mapped Out

1. Daily Standup Summary & Progress Gauge (~25% Completed)

Team Standup Speaking Script:

'Hello team! Here is our current project status for CyberQuant:

- **Overall Progress:** We are at **~25% completion**. The entire data layer and foundational architecture are 100% complete and live on GitHub.
- **What is Done:** We generated a realistic 150-asset enterprise dataset (total valuation: **₹39.37 Crores**) with 69 CVE vulnerabilities and 6 security controls mapped to RBI, NIST, and ISO standards. The repository is fully structured.
- **What is Next:** Next is **Milestone 2 (Risk Quantification Engine)** to calculate Expected Annual Loss (EAL) in Rupees, followed by the FastAPI web server and the Knapsack budget optimizer.'

2. What We Have Done So Far (Milestone 1: 100% Completed)

Milestone 1 establishes the foundational data architecture for the entire platform. Every component below is verified and committed to GitHub:

Deliverable	Technical Description	Key Metrics / Outputs	Status
generate_data.py	Deterministic Python data generator with random.seed(42). Simulates enterprise telemetry without violating bank confidentiality.	Produces 5 interconnected JSON datasets in under 1 second.	DONE ■
assets.json	Inventory of 150 enterprise computers across Critical, High, Medium, and Low tiers with financial valuation in ■ and downtime rates.	150 Assets Total Valuation: ₹39.37 Crores (Critical: Core Banking DB, Payment API, IAM).	DONE ■
vulnerabilities.json	Real-world CVE flaws mapped across assets with CVSS severity (0.0–10.0), exploit availability flags, and remediation patch costs.	69 CVE Flaws (52 Critical) E.g. CVE-2024-3094, CVE-2024-21413.	DONE ■
controls.json	6 enterprise defense controls with implementation costs, risk reduction metrics, and regulatory framework tags.	MFA, Patch Program, EDR, Network Wall, Cloud WAF, Backups Mapped to RBI, SEBI, NIST CSF, ISO 27001.	DONE ■
threats.json & summary.json	Active threat intelligence vectors (Ransomware, Credential Stuffing, API Abuse) and enterprise metadata summary.	5 Threat Scenarios with annual baseline likelihood percentages.	DONE ■
GitHub Repository Setup	Repository initialized, branch set to main, with requirements.txt, .gitignore, and comprehensive README.md.	Live at: https://github.com/laksh1441942-wq/CyberQuant	DONE ■

3. What is Left to Do (Milestones 2 to 6)

Here is the detailed breakdown of the remaining 5 milestones required to complete CyberQuant for the internal hackathon demo:

Milestone & Module	Core Objectives & What Needs to be Built	Expected Key Output	Estimated Effort
Milestone 2: Risk Engine <code>risk_engine.py</code>	Implement the FAIR (Factor Analysis of Information Risk) mathematical model. Calculate incident likelihood from CVSS and exposure, compute financial impact in ₹ , and evaluate Expected Annual Loss (\$ ext{EAL} = P imes I\$) for every asset and the enterprise.	Enterprise EAL: ₹12.05 Crores , Enterprise Risk Score (41/100), and Top 5 Critical Financial Risks.	Day 2 <i>(Next Step)</i>
Milestone 3: FastAPI Web Server <code>main.py</code>	Build the asynchronous Python web server using FastAPI and Uvicorn . Enable CORS middleware. Implement REST endpoints: GET <code>/api/dashboard</code> (summary stats) and GET <code>/api/assets</code> (150 evaluated assets).	Live interactive Swagger documentation screen running at <code>http://127.0.0.1:8000/docs</code> .	Day 3
Milestone 4: Budget Optimizer & Scenarios <code>optimizer.py</code>	Implement the Greedy Knapsack Algorithm : endpoint POST <code>/api/investment/optimize</code> taking a budget (e.g. ₹1 Crore) and finding the combination of controls delivering maximum risk reduction and ROSI %. Add What-If simulation route POST <code>/api/scenario</code> .	Automatic optimal spend recommendation (e.g. spending ₹1 Cr saves ₹1.2 Cr in expected loss; ROSI = 20%).	Day 4
Milestone 5: Frontend Dashboard UI <code>index.html</code> , <code>Chart.js</code>	Develop a clean, dark-mode glassmorphic dashboard in HTML/CSS/JavaScript. Integrate Chart.js for risk score dials, asset donut charts, and loss trend lines. Add dynamic Budget Slider and Offline Mock Switch (MOCK_MODE=True) .	Interactive browser dashboard that updates live when users slide the budget or test scenarios.	Day 5
Milestone 6: Demo Rehearsal & Pitch Deck Final Presentation	Prepare the winning 6-minute live demonstration flow. Create slide deck covering Problem, FAIR Model, Architecture, and Business ROSI. Rehearse 5 times with offline backup screenshots.	Rock-solid, rehearsed presentation ready for SIH evaluators with zero chance of crashes.	Day 6

4. How We Are Going to Execute the Remaining Work

To ensure zero delays, zero budget expenditure, and rock-solid code quality, we follow a strict 4-step development discipline for each milestone:

- **Step 1 (100% Free Open-Source Tools):** We use Python, FastAPI, SQLite, Pandas, and Chart.js. No paid cloud services or subscriptions are required; everything runs locally on a single laptop.
- **Step 2 (Local Verification First):** Every module (Math Engine, FastAPI endpoints, Knapsack algorithm) is written and tested locally with assertion tests before being merged.
- **Step 3 (Continuous GitHub Commits):** Each completed milestone will be committed with a clean conventional commit (e.g. feat: Day 2 - Risk Quantification Engine) and pushed to origin/main.
- **Step 4 (Failsafe Live Demo Protocol):** The backend will include a `MOCK_MODE = True` toggle so that if the venue network drops during the presentation on September 2, the system seamlessly serves cached results without crashing.

5. Immediate Next Action Item

The immediate next task is **Milestone 2: Finalizing and committing `risk_engine.py`** to the repository. This will transform our 150 synthetic assets into quantified financial risk (Expected Annual Loss in **₹**) and unblock the backend API routes.